

Proyecto Tesis

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1 State of the art

1.1 Differential Privacy

Differential privacy describes a promise that is made by a data holder or curator to a data subject: "You will not be affected, adversely or otherwise, by allowing your data to be used in any study or analysis, no matter what other studies, data sets, or information sources, are available".[1] In other words differential privacy is the process that either a data curator or a data holder must do in order to assure that a single individual's data can be extracted from the study as a whole.

Definition 1.1. *(ϵ, δ) -Differential privacy [1]. A randomized algorithm $\mathcal{M}$ will provide (ϵ, δ) -Differential privacy (henceforth (ϵ, δ) -DP) if for any two outputs on two different databases D and D' differing by at most one record, and for all outputs $A \subseteq \text{Range}(O)$:*

$$\Pr[\mathcal{M}(D) \in O] \leq e^\epsilon \Pr[\mathcal{M}(D') \in O]$$

In this study we are going to mostly focus around the concept of **Central differential privacy** which states that it is the responsibility of the curator (i.e a central party that receives all the raw data from the participants and adds privacy to them) to assure that the data is going to be differentially private.

1.2 Quantum Information and Computation

Quantum computing significantly advances computational capabilities, performing exponentially faster computation than classical computers.[2] In order to understand some concepts of Quantum differential privacy it is important to define the following notions:

Definition 1.2. *Quantum states: A quantum state of the system X is described by a density matrix ρ whose entries are complex numbers and whose indices (for both its rows and columns) have been placed in correspondence with the classical state set Σ*

Definition 1.3. *Mixed state: A mixed state of a quantum system is represented by an ensemble*

$$\mathbb{E} = (p_1, |\psi_1\rangle), \dots, (p_k, |\psi_k\rangle)$$

which in turn can be represented by the density operator

$$\rho_{\mathbb{E}} = \sum_i p_i |\psi_i\rangle \langle \psi_i|$$

thus the resulting $\rho_{\mathbb{E}}$ will be the density matrix representation of the ensemble.

Definition 1.4. *Quantum Channel:* A quantum channel describes a linear mapping over density matrices.

Quantum channels are important as they are the generalization of an operation over a quantum state. This generalization is necessary as in quantum differential privacy the noise introduced into the system will be destructive over the quantum state and thus not representative of a more specific definition of operation

Definition 1.5. *Trace distance:* It is the distance between two density operators and it is defined by:

$$\tau(\rho, \sigma) = \frac{1}{2} \text{Tr}|\rho - \sigma|$$

where $\text{Tr}(x)$ is the trace of the density matrix x

Definition 1.6. *Quantum measurement:* It is the way to extract information from a quantum system. Mathematically, a measurement is modelled as a set of matrices $M = K_m$ with the condition of

$$\sum_m K_m^\dagger K_m = I$$

with I being the identity of the density operator (i.e for a qbit $I = 2 \times 2$ identity matrix)

when talking about quantum differential privacy below it won't really matter what the quantum state collapses into thus we can use the notion of a POVM (Positive Operator-Valued Measure) which is a simplification over the measurement where we get that $M = M_m$ with $M_m = K_m^\dagger K_m$ $\forall m$ since $p_m = \text{Tr}(K_m \rho K_m^\dagger) = \text{Tr}(M_m \rho)$

1.3 Quantum Differential Privacy

When understanding classical differential privacy one might realize the possibility for it to be applied as well to quantum information. The promise that classical DP curator makes to the data holder does not change when in the quantum setting however the means to reach that promise do

Definition 1.7. *(Quantum differential privacy [1]).* Let $d \in (0, 1]$ and $\epsilon, \delta > 0$ be three constants. A quantum operation $\mathcal{E}$ is (ϵ, δ) -differentially private if for every POVM $M = M_m$, for all outputs $S \subseteq \text{Out}(M)$ and for all inputs ρ, σ such that $\tau(\rho, \sigma) \leq d$, it holds that:

$$\Pr[\mathcal{E}(\rho) \in_M S] \leq e^\epsilon \Pr[\mathcal{E}(\sigma) \in_M S]$$

2 Description

3 Objectives

3.1 General Objective

The main objective of this project is to empirically validate the theoretical (ϵ, δ) – DP guarantees provided by quantum noise mechanisms such as the depolarizing channel and GAD, by comparing

the theoretical privacy budget derived in [1] with the empirical privacy budget measured on a practical quantum simulation in the pennylane framework.

3.2 Specific Objectives

- Gain an understanding of the classical definition of differential privacy, as well as quantum information and computation to understand quantum differential privacy.
- Create an experiment on the quantum simulation framework pennylane which will test a simple toy example proposed in [1] as a base line.
- Create an experiment on the quantum simulation framework pennylane which will test $(\epsilon, \delta) - DP$ over a real dataset[1].
- Establish whether the theory proposed in [1] and [2] agrees or disagrees with the results of the experiments

4 Expected Results

The expected results of this project are as follows:

1. Prove that in practice the privacy budget will be preserved within reasonable proximity to the theoretical budget in an experimental setup.
2. Provide the source code for the experimental setup in pennylane with python.
3. Write a scientific article in the format of IEEE in which I document the project in appropriate manner. The information within this article should demonstrate my understanding of the topic as it will contain the whole process of investigation, creation and result analysis of this project.
4. Design a poster which will summarize the most important parts of the project.

5 Ethical Considerations

6 Activity Calendar

References

References